Cofunction identities

1 For each of the following acute angles, state the complementary angle:

a 34° **b** 62° **c** $\frac{\pi}{6}$ **d** $\frac{2\pi}{7}$

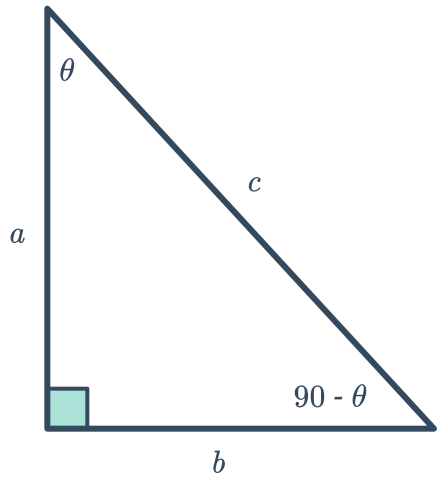
2 Write each of the following trigonometric ratios in terms of its cofunction:

a $\cos 37^\circ$ **b** $\sin 76^\circ$ **c** $\tan 68^\circ$ **d** $\cot(43^\circ)$

e $\operatorname{cosec} 35^\circ$ **f** $\cos \frac{\pi}{9}$ **g** $\sin \frac{\pi}{10}$ **h** $\cot\left(\frac{\pi}{9}\right)$

i $\sec \frac{\pi}{9}$ **j** $\cos \frac{1}{5}$

3 Consider the following right triangle:



a Find an expression for the following:

i $\cos\theta$ **ii** $\sin(90^\circ - \theta)$

iii $\sin\theta$ **iv** $\cos(90^\circ - \theta)$

b Describe the rule found in part (a) using words.

c If $\sin\alpha = 0.32$, find $\cos(90^\circ - \alpha)$.

d If $\sin\alpha = \cos\beta$, find $\alpha + \beta$.

4 Simplify:

a $\sin(90^\circ - p)$ **b** $\frac{\cos(90^\circ - x)}{\cos x}$

c $\frac{\cos 26^\circ}{7\sin 64^\circ}$ **d** $\frac{\sin 51^\circ}{\cos 39^\circ}$

e $\sin(90^\circ - y) \times \tan y$

5 Given that $\sin x = 0.19$, find the exact value of $\cos\left(\frac{\pi}{2} - x\right)$.

6 Given that $\sin x = 0.46$, find the exact value of $\cos(90^\circ - x)$.

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7 Prove the following:

a $\dfrac{\sin x \cos(90 - x)}{\cos x \sin(90 - x)} = \tan^2(x)$

b $\dfrac{\sin x \sin(90 - x)}{\cos x \cos(90 - x)} = 1$

Trigonometric equations

8 Solve for the measure of the acute angle θ in the following equations:

a $\sin\theta = \cos 25^\circ$ **b** $\cos\theta = \sin 85^\circ$ **c** $\cos\theta = \sin\dfrac{\pi}{8}$ **d** $\sin\dfrac{2\pi}{5} = \cos\theta$

9 Solve for the exact value of c, if $\sin\left(\dfrac{\pi}{9} + c\right) = \cos\left(\dfrac{\pi}{4}\right)$.

10 Solve for the value of c, if $\sin(34^\circ + c^\circ) = \cos 51^\circ$.

11 Solve for the value of B if $\cos(66^\circ - B) = \sin 38^\circ$.

12 Solve for the value of A if $\sin\left(\dfrac{A}{68}\right) = \cos 18^\circ$.

13 Solve for the value of x in the following equations:

a $\sin(5x + 40^\circ) = \cos(3x + 10^\circ)$

b $\sin\left(6x + \dfrac{\pi}{3}\right) = \cos\left(4x + \dfrac{\pi}{9}\right)$

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